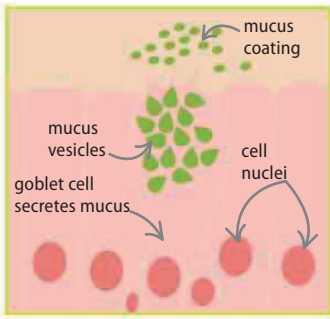


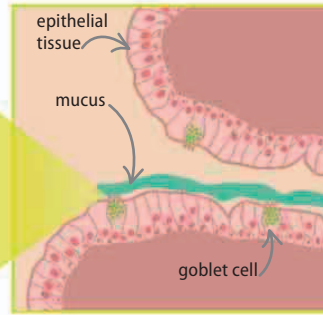
## Multicellular structures

A living body is made of billions of cells working together. To do that most effectively, the cells are specialized to do certain jobs. A collection of cells that performs a single function—such as producing the mucus in the nose—is called a tissue. Very often, tissues group together to perform a complex set of tasks. They are then described as an organ, such as the nose.



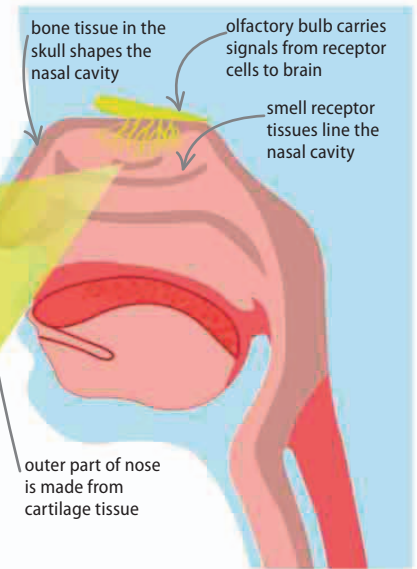
### △ Goblet cell

This type of cell produces mucus (a mixture of water and a gooey protein called mucin) and other dissolved chemicals.



### △ Epithelial tissue

Goblet cells form much of the epithelia, the tissue that lines the nose, windpipe, and gut. The mucus they produce protects the cells from chemical attack and dirt.

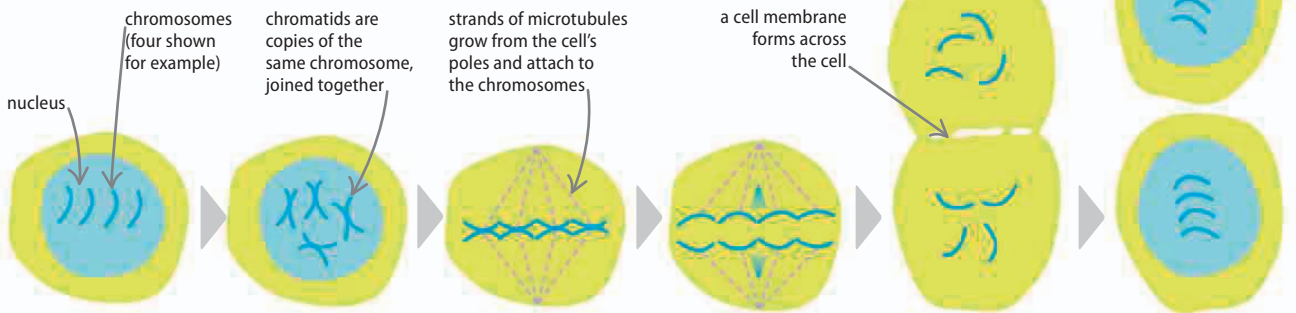


### △ Nose

The nose is an organ that carries air in and out of the body. Muscle, cartilage, and bone tissues combine with epithelial tissue to help it do its job.

## Cell division

A body grows because the number of its cells increases. This increase in number is achieved by cells dividing in half, to make two identical but fully independent cells. This type of cell division is called mitosis. It involves several stages, in which the cell's contents are split into two groups. That includes doubling the number of chromosomes (which carry the cell's genes).



### △ 1. Interphase

Cell has usual number of 46 chromosomes inside it.

### △ 2. Prophase

Each chromosome is doubled, forming two chromatids.

### △ 3. Metaphase

The chromosomes line up in the middle of the cell.

### △ 4. Anaphase

The chromatids are pulled apart, to become separate chromosomes.

### △ 5. Telophase

The microtubules disappear, and the cells begin to divide.

### △ 6. Cytokinesis

Two daughter cells are formed, each with 46 chromosomes.